Chapter 1. Introduction to Nonparametric Statistics

Nonparametric models, parametric inference review, why nonparametric methods, robustness

Contents

1.1 Parametric and nonparametric models **2**

1.2 Review of parametric inference **2**

1.3 Why nonparametric methods **3**

1.4 Selected exercises **4**

1.1 Parametric and nonparametric models

A statistical model is a family $\mathcal{F}$ of distributions assumed to contain the population. It is **parametric** if it is indexed by finitely many real numbers, e.g. the normal family

$$\mathcal{F}_G = \{N(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma > 0\}. \tag{1.1}$$

It is **nonparametric** if no finite-dimensional parametrization exists. Typical examples:

$$\mathcal{F}_1 = \{F : \int x^2\, dF < \infty\}, \qquad \mathcal{F}_{\text{all}} = \{\text{all } F\},$$
$$\mathcal{F}_2 = \{F : F \text{ has a density symmetric about some } \theta\}.$$

Both $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. A model with a finite-dimensional part of interest and an infinite-dimensional nuisance part, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F$ unrestricted, is called **semiparametric**.

In a nonparametric model a **parameter** is a functional $\theta = \theta(F)$: the mean $\int x\, dF$, the median, a quantile, the density f, or a regression curve $m(x) = E(Y \mid X = x)$. Knowing a few such values never determines F. A procedure whose validity (e.g. type I error) holds for every F in a very large family is called **distribution-free**.

The course uses three tools:

1. **Ranks, signs and permutation tests** (Part I): conditional inference given the observed values, exact under exchangeability.
2. **The bootstrap** (Part II): replace the unknown F by an estimate and compute sampling distributions by simulation.
3. **Smoothing** (Part II): estimate a function $f(x)$ or $m(x)$ by averaging the observations near x.

1.2 Review of parametric inference

Motivation. Nonparametric methods are best understood as answers to the failures of parametric ones. We recall the t-test and the likelihood framework so that later we can see exactly which assumptions each rank method drops and what it costs in efficiency.

Definition (Quantile)

For a distribution function F and $p \in (0,1)$, the p-quantile is $Q(p) = \min\{x : F(x) \geq p\}$. For the standard normal, $z_\alpha := \Phi^{-1}(1-\alpha)$.

Proposition 1.1

Let F be the distribution function of an integer-valued X (e.g. $X \sim B(n,p)$). Then $q = Q(p)$ is the smallest integer with $F(q) \geq p$, and $F(q-1) < p \leq F(q)$. In particular $P(X \leq q) \geq p$ and $P(X \geq q) > 1 - p$.

Proof. F is nondecreasing, right-continuous and constant on each $[k, k+1)$, so $\{x : F(x) \geq p\} = [q, \infty)$ with q the smallest integer satisfying $F(q) \geq p$. Minimality gives $F(q-1) < p$. Then $P(X \geq q) = 1 - F(q-1) > 1 - p$. ■

Two reference distributions recur. If $X \sim N(\mu, \sigma^2)$ then $(X-\mu)/\sigma \sim N(0,1)$, so $F_X(x) = \Phi((x-\mu)/\sigma)$. If $X \sim B(n,p)$ (number of successes in n independent trials), then $EX = np$, $\operatorname{Var} X = np(1-p)$ (a sum of n Bernoulli(p) variables), and for large n (roughly $np \geq 10$) X is approximately $N(np, np(1-p))$ by the CLT below.

Estimation. An estimator $\hat{\theta}$ is judged by its bias $E\hat{\theta} - \theta$, standard error $\mathrm{se}(\hat{\theta}) = \sqrt{\mathrm{Var}\,\hat{\theta}}$ and mean squared error

$$\mathrm{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2 = \mathrm{bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Under $\mathcal{F}_G$, $\bar{X}_n$ is the best unbiased estimator of μ, and $T = \sqrt{n}(\bar{X}_n - \mu)/s \sim t_{n-1}$ gives the exact interval $\bar{X}_n \pm t_{\alpha/2,n-1}s/\sqrt{n}$. Outside $\mathcal{F}_G$ these optimality and exactness statements are lost.

Testing. A test of H_0 against H_1 rejects when a statistic T falls in a rejection region chosen so that $P_{H_0}(\text{reject}) \le \alpha$ (the **level**); among such tests we want large **power** $P_{H_1}(\text{reject})$. The **p-value** is the probability under H_0 of a value of T at least as extreme (in the direction of H_1) as the observed one; rejecting when $p \le \alpha$ is the same as using the level-α rejection region. In nonparametric problems the central task is to find a statistic whose **null distribution** does not depend on the unknown F.

Proposition 1.2 (Duality of tests and intervals)

Suppose for each θ_0 there is a level-α test ϕ_{θ_0} of $H_0 : \theta = \theta_0$. Then $C = \{\theta_0 : \phi_{\theta_0} \text{ does not reject}\}$ is a $1-\alpha$ confidence set. Conversely, if C is a $1-\alpha$ confidence set, rejecting $H_0 : \theta = \theta_0$ when $\theta_0 \notin C$ is a level-α test.

Proof. If θ is the true value, $P(\theta \in C) = P(\phi_\theta \text{ does not reject}) \ge 1-\alpha$. Conversely $P_{\theta_0}(\theta_0 \notin C) \le \alpha$. ■

The central limit theorem. Much of classical practice relies on large-sample normality.

Theorem 1.3 (CLT)

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with mean μ and variance $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof [Supplement]. Let $Y_i = (X_i - \mu)/\sigma$ and $\varphi(t) = Ee^{itY_1}$. Since $EY_1^2 < \infty$, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By independence $Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(\frac{1}{n})\right)^n \to e^{-t^2/2}$, the characteristic function of $N(0,1)$; Lévy's continuity theorem concludes. ■

Because of the CLT, t-based procedures are routinely applied to discrete, ordinal or skewed data: what matters is that the *statistic*, not the population, is approximately normal. The CLT says nothing about how large n must be. Convergence is slow for skewed F and fails entirely when $\sigma^2 = \infty$ (e.g. Cauchy or slash populations). This is the first motivation for nonparametric methods.

1.3 Why nonparametric methods

- *Validity under weak assumptions.* Rank, sign and permutation tests control the type I error exactly for every F in a large family, at any sample size.
- *Computation instead of derivation.* Permutation and bootstrap distributions replace formulas that are unknown or valid only asymptotically.
- *Flexibility.* Function-valued targets (f, m) are estimated without assuming a shape.
- *Robustness.* Procedures based on ranks, signs or medians are insensitive to outliers.

Example 1.1 (Effect of one outlier)

Take the 8 values 2.1, 2.4, 2.6, 2.9, 3.0, 3.2, 3.5, 25. With the last value, the mean is 5.59 and the standard deviation 7.86; without it, 2.81 and 0.48. The median is 2.95 with it and 2.9

without. The **median absolute deviation** MAD $= \text{median}_i |x_i - \text{median}_j\, x_j|$ is 0.45 with it and 0.30 without. The mean and standard deviation are dominated by a single point; the median and MAD barely move.

An outlier can be viewed as a draw from a heavy-tailed population. Distribution-free methods keep their properties under heavy tails, so they are typically robust (the converse need not hold).

1.4 Selected exercises

Motivation. These textbook exercises practise the parametric tools reviewed above (quantiles, CLT-based intervals and tests) and show how an exact pivot, when the model provides one, replaces the normal approximation.

Exercise 1.1 (Normal quantiles)

Express the quantile function $Q(p)$ of $N(\mu, \sigma^2)$ through Φ, and find the median $Q(0.5)$ and the quartiles $Q(0.25)$, $Q(0.75)$.

Solution.

Step 1. $F(x) = \Phi((x-\mu)/\sigma)$ is continuous and strictly increasing on $\mathbb{R}$ with limits 0 and 1. So for $p \in (0,1)$ there is a unique x_p with $F(x_p) = p$, and $\{x : F(x) \geq p\} = [x_p, \infty)$. Hence $Q(p) = x_p$.

Step 2. $F(x_p) = p$ means $(x_p - \mu)/\sigma = \Phi^{-1}(p)$, so

$$Q(p) = \mu + \sigma\, \Phi^{-1}(p).$$

Step 3. Since ϕ is even, $\Phi(-z) = 1 - \Phi(z)$, so $\Phi^{-1}(1-p) = -\Phi^{-1}(p)$ and $\Phi^{-1}(0.5) = 0$. With $\Phi^{-1}(0.75) = 0.6745$,

$$Q(0.5) = \mu, \qquad Q(0.25) = \mu - 0.6745\,\sigma, \qquad Q(0.75) = \mu + 0.6745\,\sigma.$$

In particular the interquartile range of a normal population is $1.349\,\sigma$. This is the source of the divisor 1.34 in the bandwidth rule of Section 11.2, and of the factor $1/0.6745 = 1.4826$ that turns the MAD into an estimate of σ (Section 10.3.1). ■

Exercise 1.5 (Exponential waiting times)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\theta)$ with mean $\theta > 0$, i.e. density $\theta^{-1} e^{-x/\theta}$ for $x > 0$.
(a) Use the CLT to find the approximate distribution of $\bar{X}_n$.
(b) The waiting time for a bus is modeled this way, and $n = 40$ students waited $\bar{x} = 7$ minutes on average. Give an approximate 95% confidence interval for θ, and test $H_0 : \theta = 10$ against $H_1 : \theta < 10$; report the p-value and the conclusion at $\alpha = 0.01$.

Solution.

(a) With $y = x/\theta$, $EX = \theta \int_0^\infty y e^{-y}\, dy = \theta$ and $EX^2 = \theta^2 \int_0^\infty y^2 e^{-y}\, dy = 2\theta^2$, so $\text{Var}\, X = \theta^2$. By Theorem 1.3, $\sqrt{n}(\bar{X}_n - \theta)/\theta \xrightarrow{d} N(0,1)$, i.e. $\bar{X}_n$ is approximately $N(\theta, \theta^2/n)$.

(b) *Interval.* By the law of large numbers $\bar{X}_n/\theta \xrightarrow{p} 1$, so by Slutsky's lemma $\sqrt{n}(\bar{X}_n - \theta)/\bar{X}_n \xrightarrow{d} N(0,1)$ as well. This gives the interval

$$\bar{x} \pm 1.96\, \frac{\bar{x}}{\sqrt{n}} = 7 \pm 1.96 \cdot \frac{7}{\sqrt{40}} = 7 \pm 2.17 = (4.83,\ 9.17).$$

Test. Under H_0 the standard deviation of $\bar{X}_n$ is known, $\theta_0/\sqrt{n}$, so we use $Z = \sqrt{n}(\bar{X}_n - \theta_0)/\theta_0$ and reject for small Z. Here $z = \sqrt{40}(7-10)/10 = -1.90$ and $p = \Phi(-1.90) = 0.029$. Since $p > 0.01$, H_0 is not rejected at the 1% level (it would be at 5%). ■

Exercise 1.7 (Exact tests under a model)

Can the test of Exercise 1.5 be carried out exactly, without the CLT, if the population is assumed to be (a) Exp(θ), or (b) Uniform$(0, 2\theta)$ (which also has mean θ)?

Solution.

(a)

Step 1. (A pivot.) $Y_i = 2X_i/\theta$ has density $\frac{1}{2}e^{-y/2}$, $y > 0$, which is the χ^2_2 density. The moment generating function of χ^2_2 is $(1-2t)^{-1}$ for $t < \frac{1}{2}$, so by independence $\sum_i Y_i$ has m.g.f. $(1-2t)^{-n}$, that of χ^2_{2n}. Hence

$$V(\theta) = \frac{2n\bar{X}_n}{\theta} \sim \chi^2_{2n} \qquad \text{for every } \theta > 0.$$

Step 2. (Test.) Under H_0, $V(\theta_0) \sim \chi^2_{2n}$. If the true value is $\theta < \theta_0$, then $V(\theta_0) = (\theta/\theta_0)V(\theta)$ is stochastically smaller, so we reject for small $V(\theta_0)$ and $p = P(\chi^2_{2n} \le 2n\bar{x}/\theta_0)$. Here $v = 2{\cdot}40{\cdot}7/10 = 56$ and $p = P(\chi^2_{80} \le 56) = 0.019$, again not significant at 1%.

Step 3. (Interval.) With q_p the p-quantile of χ^2_{2n}, $P(q_{0.025} \le V(\theta) \le q_{0.975}) = 0.95$, i.e. $\theta \in (2n\bar{X}_n/q_{0.975},\ 2n\bar{X}_n/q_{0.025})$ with probability exactly 0.95. Numerically this is (5.25, 9.80). Compared with Exercise 1.5, the exact interval is shifted to the right and the exact p-value is smaller: the distribution of $\bar{X}_n$ is right-skewed, which the normal approximation ignores.

(b)

Step 1. (A pivot.) Let $M = \max_i X_i$. For $0 \le m \le 2\theta$, $P(M \le m) = \prod_i P(X_i \le m) = (m/2\theta)^n$. Hence $U(\theta) = (M/2\theta)^n$ satisfies $P(U(\theta) \le u) = P(M \le 2\theta u^{1/n}) = u$ for $u \in [0,1]$, i.e. $U(\theta) \sim$ Uniform$(0,1)$ for every θ.

Step 2. (Test.) Under $\theta < \theta_0$ all observations lie below $2\theta < 2\theta_0$, so small M supports H_1. The p-value is $p = P_{\theta_0}(M \le m_{\text{obs}}) = \min\{1, (m_{\text{obs}}/2\theta_0)^n\}$. Under H_0, $p = U(\theta_0)$ is exactly uniform, so $P(p \le \alpha) = \alpha$: the test is exact.

So in both models an exact test exists, but it uses a model-specific statistic ($\bar{X}_n$ with the χ^2 law, or the maximum). The summary $\bar{x} = 7$ alone is not enough for (b). Each test is exact only under its own model; this dependence on the model is what nonparametric methods avoid. ■